A380X cockpit buttons, lights and probes

Dead A380X cockpit pushbuttons now work, the 52 RESET panel buttons can be clicked, the FMC-C reset button is checked, the AUTOLAND lights flash, the side slip and angle of attack vanes on the nose move, the ACD keys beside the OITs work and the F/O laptop keyboard has its keys

Aircraft	A380X
Where to find it	Thrust levers, overhead EVAC and RESET panels, right pedestal, glareshield AUTOLAND lights, nose probes (outside view)
Branch	`core/a380x-cockpit`, based on `master` (FlyByWire master 76db0aef1 merged in); independent of the other core branches. The MFP vanes and the ACD / F/O keyboard also need their companion aircraft-large-files changes (fork TCdr/aircraft-large-files, branches feat/a380x-mfp-vane-animations, b8fb739, and feat/a380x-acd-keys-fo-keyboard, 7f56bcb).
Status	All six features tested in the sim by the user: OK. Pushbuttons, RESET panel and side slip vanes, FMC-C reset check; AUTOLAND light and MFP vanes; ACD keys and F/O keyboard.
Pictures	No pictures: the buttons and the vanes are the existing cockpit and fuselage models.

1. A/THR disconnect, EVAC HORN, DFDR EVENT and ACMS TRIGGER pushbuttons work
2. Clickable RESET panel buttons and side slip probe vanes
3. FMC-C reset button checked in the all-FMCs-reset test
4. AUTOLAND light press-to-test and flashing
5. MFP vanes follow the angle of attack
6. ACD keys beside the OITs work; F/O laptop keyboard keys

1. A/THR disconnect, EVAC HORN, DFDR EVENT and ACMS TRIGGER pushbuttons work

> **Aircraft:** A380X | **Where:** Thrust levers 1 and 4, overhead EVAC panel, right pedestal | **Status:** Tested in the sim by the user: OK.

What it is for

The cockpit behaviour XML named model nodes or animations that are not in the A380X cockpit model, so these pushbuttons could not be clicked or did not move. Checked against the cockpit model:

Pushbutton	Before	Now
A/THR instinctive disconnect pbs (thrust levers 1 and 4)	Not clickable (PUSH_THROTTLE_1/2)	Clickable (ATHR_DISCONNECT_ENG1/2): they disconnect the A/THR through the existing FBW interface
EVAC HORN SHUT OFF	Not clickable (PUSH_OVHD_EVAC_HORN)	Clickable (PUSH_OVHD_EVAC_HORN_OFF)
DFDR EVENT, ACMS TRIGGER (right pedestal)	Did not move	Move when pressed (ANIM_PUSH_DFDR_EVENT / ANIM_PUSH_ACMS_TRIGGER)

Limitations and why

Limitation	Why	What to do
DFDR EVENT and ACMS TRIGGER only move.	There is no flight data recorder or ACMS function behind them; their tooltips still say (Inop.).	None.

How to use

1. Press an A/THR instinctive disconnect pushbutton on thrust lever 1 or 4 to disconnect the A/THR.
2. Press EVAC HORN SHUT OFF on the overhead EVAC panel, DFDR EVENT or ACMS TRIGGER on the right pedestal as on the aircraft.

How to test

#	Do this	Expected result
1	A380X in flight, A/THR active. Click the instinctive disconnect pushbutton on thrust lever 1.	The pushbutton moves and the A/THR disconnects (A/THR leaves the FMA).
2	Engage the A/THR again and click the pushbutton on thrust lever 4.	Same result.
3	Click EVAC HORN SHUT OFF on the overhead EVAC panel.	The pushbutton can be clicked and moves.
4	Click DFDR EVENT and ACMS TRIGGER on the right pedestal.	Each one moves in and out when pressed.

2. Clickable RESET panel buttons and side slip probe vanes

> **Aircraft:** A380X | **Where:** Overhead RESET panels; nose side slip probes | **Status:** Tested in the sim by the user: OK; the user accepted that 42 buttons only drive their variable, and confirmed the vane direction.

What it is for

All 52 circuit breakers of the overhead RESET panels can now be pulled and pushed, with their FCOM label as tooltip. Each one writes `L:A32NX_RESET_PANEL_<name>` (true = pulled, computer reset).

- Read by the avionics: ARPT NAV, FMC A, FMC B, FMC C, FWS 1, FWS 2, AESU 1, AESU 2, NSS AVNCS, NSS FLT OPS.
- The other 42 (AICU, ATC, AVS, BSCS, CIDS, CPCS, DSMS, DTLNK ROUTER, EIPMs, ESS TR, TR 1, TR 2A, FLAPS, FQMS, GCU, LGCIS, PACK CTL, PAX BBAND, SCS, SDF, SLATS, TCS, VCS) are clickable but no system reads them yet.

The three side slip probe vanes on the nose now follow the side slip angle (sim `INCIDENCE BETA`, up to 60 degrees each side): they line up with the airflow.

Limitations and why

Limitation	Why	What to do
42 RESET buttons do nothing in the systems.	The systems they reset are not wired to the RESET panel yet.	Use the 10 wired ones to test a reset.

How to use

1. Hover a RESET panel button to read its label; click to pull it (reset), click again to push it back.
2. Look at the nose from the outside view to see the side slip vanes.

How to test

#	Do this	Expected result
1	Hover the buttons of the overhead RESET panels.	Each one shows its FCOM label as tooltip.
2	Click a button.	It pulls out; click again: it pushes in.
3	Click one of the 42 not wired buttons, for example a PACK CTL button.	It moves; no system reacts.
4	In flight, outside view on the nose. Fly a steady sideslip with the rudder (for example right rudder).	The three side slip vanes turn to line up with the airflow; centred in coordinated flight.

3. FMC-C reset button checked in the all-FMCs-reset test

> **Aircraft:** A380X | **Where:** Overhead RESET panels: FMC A, FMC B, FMC C | **Status:** Tested in the sim by the user: OK: FMC-A and FMC-B pulled, the FMS kept the plan with FMC-C in. Unit tests: FmcResetPanel.spec.ts.

What it is for

The FMS stops and resets when all the FMC reset buttons are pulled. The check read FMC-B twice and never FMC-C, so pulling FMC-A and FMC-B alone reset the FMS and lost the flight plan while FMC-C was still in. Per A380 FCOM DSC-22-FMS-30 FMC-C is the standby FMC, and FMS 1 and FMS 2 are only lost when FMC-A, FMC-B and FMC-C all fail (PRO-ABN-ECAM-10-22-FMS AUTO FLT FMS 1+2 FAULT). The check now covers A, B and C.

What you should see

Situation	What you should see
FMC-A and FMC-B pulled, FMC-C in	The FMS keeps running; the flight plan stays.
FMC-A, FMC-B and FMC-C pulled	The FMS stops and resets (the flight plan is lost), as before.

How to use

1. Pull FMC A, FMC B and / or FMC C on the overhead RESET panels to reset those FMCs; push them back in to restore.

How to test

#	Do this	Expected result
1	A380X with a flight plan in the MFD F-PLN page.	The plan is shown.
2	Pull the **FMC A** and **FMC B** reset buttons.	The flight plan stays on the MFD (FMC-C still in).
3	Pull **FMC C** as well.	The FMS resets: the flight plan is lost.

4. AUTOLAND light press-to-test and flashing

> **Aircraft:** A380X | **Where:** Glareshield, captain and F/O AUTOLAND lights | **Status:** Tested in the sim by the user: OK.

What it is for

A380 FCOM DSC-22-FG-80-100: the AUTOLAND light flashes in autoland below 200 ft RA when the autoland conditions are lost, and "as long as the AUTOLAND light is pressed, the AUTOLAND light flashes" (light test). The cockpit lamp code cannot read the sim time, so the old flashing expression never lit the light. The flashing (1 Hz: half on, half off) is now computed in the systems host, for the light test and for the autoland warning.

What you should see

Situation	What you should see
AUTOLAND light held pressed	It flashes red at 1 Hz until released.
Autoland below 200 ft RA, autoland conditions lost	Both AUTOLAND lights flash (the warning itself is the existing FBW logic).

How to use

1. Press and hold the AUTOLAND light on the glareshield to test it (cockpit preparation).

How to test

#	Do this	Expected result
1	A380X powered, on the ground. Press and hold the captain AUTOLAND light.	It flashes red, about once per second.
2	Release it.	It goes off.
3	Repeat with the F/O AUTOLAND light.	Same.

5. MFP vanes follow the angle of attack

> **Aircraft:** A380X | **Where:** Nose multifunction probes (outside view) | **Status:** Tested in the sim by the user: OK.

What it is for

The angle of attack vanes of the three multifunction probes on the nose (one on the left, two on the right, A380 FCOM DSC-34-10-30-10) now follow the angle of attack (sim `INCIDENCE ALPHA`, up to 60 degrees each way, smoothed so that the jumps at very low speed on the ground do not show).

Limitations and why

Limitation	Why	What to do
The vanes do not move without the companion model change.	The MFP_ANGLE1..3 animations are in the A380_FUSELAGE model of the aircraft-large-files repository (fork branch feat/a380x-mfp-vane-animations, b8fb739).	Install that model change with this branch.

How to use

1. Look at the nose probes from the outside view in flight.

How to test

#	Do this	Expected result
1	A380X in flight, outside view on the nose probes. Pitch up slowly to raise the angle of attack.	The three MFP vanes turn: their trailing tip rises with the angle of attack.
2	Lower the nose to a small angle of attack.	The vanes come back near their neutral position.

6. ACD keys beside the OITs work; F/O laptop keyboard keys

> **Aircraft:** A380X | **Where:** Captain and F/O side consoles: the 8-key Additional Control Device below each OIT; F/O laptop keyboard | **Status:** Tested in the sim by the user: OK: F/O laptop keyboard, all 8 captain ACD keys hover and click with the tiller centred (recorded), OIS XFR and OIS MENU. Unit tests: LaptopKeyboard.spec.ts.

What it is for

A380 FCOM DSC-46-20-20-60: each pilot has an Additional Control Device (ACD), a pad with OIS XFR, OIS MENU, ESC, ENTER and four arrow keys, for the OIT when the keyboard is stowed. Its keys were in the model but could not be clicked. They now send the same event as the laptop key of the same name, so the OIT takes ESC, ENTER and the arrows like keyboard keys, and OIS XFR / OIS MENU like the laptop OIS keys. Their legends were painted into the side console texture at about 3.5 texels per mm (a smear): each key now has a sharp, lit legend. The F/O laptop keyboard had no clickable keys and no legends: it now has the same keys, legends and press animation as the captain keyboard.

What you should see

Situation	What you should see
Mouse over an ACD key	The key highlights and shows its name (ESC, ENTER, Up, Down, Left, Right, OIS MENU, OIS XFR).
ACD key clicked	The key moves in and out; the OIT gets the key.
F/O laptop keyboard pulled out	The same legends as the captain keyboard; the keys move when clicked.

Limitations and why

Limitation	Why	What to do
OIS XFR and OIS MENU do nothing on this branch.	The OIT handling of those two keys (X-OIS, FLT OPS menu) is on core/fcom-gap-fixes.	Merge core/fcom-gap-fixes too.
The legends, click areas, F/O keyboard keys and tiller fix need the companion model change.	They are in the A380_COCKPIT_LOD00 model and two new textures of the aircraft-large-files repository (fork branch feat/a380x-acd-keys-fo-keyboard, 7f56bcb).	Install that model change with this branch.
Some keys can be hard to reach with the mouse from some head positions.	The flyPad (when mounted) and the sidestick stand between the seat and the pad. The tiller handle was split into small pieces so that it no longer hides the OIS and ENTER keys.	Move the view slightly.

How to use

1. Click the ACD keys below your OIT as on the aircraft: ENTER validates, ESC cancels, the arrows move, OIS MENU shows the FLT OPS menu, OIS XFR shows the other OIT (FLT OPS side, laptop on, logged in).
2. Pull out the F/O meal table, then the F/O keyboard, to use the F/O laptop keyboard.

How to test

#	Do this	Expected result
1	A380X powered. Pull out the F/O meal table, then the F/O keyboard; compare with the captain keyboard.	Same legends on both; F/O keys move when clicked.
2	Tiller centred. Move the mouse over each key of the captain ACD.	Each key highlights and shows its name.
3	Click ESC, ENTER and each arrow.	Each key moves.
4	With the companion OIT change (core/fcom-gap-fixes): F/O OIT on FLT OPS STS, captain OIT on the FLT OPS menu, both logged in. Click OIS XFR on the captain ACD.	The captain OIT shows the F/O STATUS PAGE and "X-OIS Activated"; OIS XFR again goes back.